Bayesian Optimisation of a Bio-Based Biodegradable Polyester Thermoset

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Abstract

Glass fibre-reinforced polymer composites are widely used in the wind turbine manufacturing industry, and are commonly manufactured using general-purpose unsaturated polyester resin thermosets. However, these resins are derived from oil-derived aromatic monomers making them both unsustainably sourced and hard to recycle. Bio-based and biodegradable alternatives exist but have not seen widespread commercial success due to their relatively high cost and poor mechanical properties, this paper seeks to address the question of whether bio-based branching thermoset polyesters can be optimised using Bayesian optimisation such that desirable properties such as extent of reaction might be improved upon. This is achieved by identifying a bio-based and biodegradable polymer (poly(glycerol citrate itaconate)), identifying a desirable objective variable (mass loss under esterification as proxy for polymerised content) and some tunable predictor variables (monomeric stoichiometry), after which sequential Bayesian techniques have been used to sample-efficiently uncover well-performing samples. A Bayesian technique employing the 'probability of improvement' acquisition function discovered the experimental global optimum stoichiometry producing a mass loss of 20.77 % in curing polyester samples, this constituted 72 % of the theoretical maximum, indicating a high extent of polymerisation. Meanwhile, an alternate sequential Bayesian technique deploying the 'expected improvement' acquisition function displayed high sample efficiency, outperforming the strongest benchmark set by a one-shot grid sampler in 63% of the samples. Both these sequential Bayesian techniques were simulated 1,000 times by querying an emulation of the practical experiment, respectively exhibiting a probability of discovering the experimental global optimum of 1.1% and 5.3%. Bayesian optimisation techniques represent a practically deployable and sample-efficient means by which desirable properties of bio-based thermesets may be improved.

1 Introduction

The early 21st century has seen continued growth in both the global population (6.17 → 8.09 billion) and total energy supply required to sustain it (416 → 634 EJ) between 2000 and 2023 [12, 30, 20]. In spite of continued strong reliance upon coal, natural gas, and oil for the provision of this demand, concerns regarding global climate and the finite nature of fossil reserves have galvanised investment in alternative renewable power systems [20, 11]. Of these renewable, wind power saw the greatest growth in capacity to generate electricity (+2.33 PW h) ahead of both hydroelectric (+1.69 PW h) and solar photovoltaics (+1.60 PW h) [19]. Wind turbine blades are commonly manufactured from a composite comprised of thermosetting polymer matrix and reinforcing material, these strong thermoset matrices are chemically cross-linked which makes them difficult to reprocess or remould [26]. Given the relatively low rate of recycling and a common rate of replacement (ca. every fifteen years in China [8]) the industry generates a large volume of practically unrecyclable thermoset waste which contravenes

the UN's 12th SDG to "ensure sustainable consumption and production" [39]. In addressing the industry's sustainability shortcoming identified above, the literature is divided into two groups: firstly, papers which focus upon improving the recyclability of contemporary materials, whether by thermal, chemical, mechanical or high-voltage fragmentation methods [40, 5, 9]; and secondly, papers aimed at the development of properties in future replacement materials that would facilitate recycling [25], such as introduction of chemically cleavable bonds into the thermoset polymers used [37]. The motivation for present work was to ascertain whether Bayesian optimisation could be used to sample efficiently improve the degree of polymerisation in a novel biodegradable bio-based thermoset.

The composite matrix used for wind turbine blades is comprised of a moulding thermoset, with the most prevalently manufactured pre-thermoset form being general-purpose unsaturated polyester resin (GP-UPR) [6]. These GP-UPRs are based on a mixture of a reactive diluent (e.g. styrene) and a linear aromatic unsaturated polyester (e.g. poly(propylene glycol maleate phthalate)), which are cured chemically or thermally to generate the final thermosetting matrix [1]. Although polyesters are known to exhibit limited biodegradability (due to their labile ester bonds being prone to both biotic and abiotic hydrolysis [34]) it is the case that polyesters with aromatic moieties are more resistant than aliphatic counterparts to such degradation [29]. Consequently, thermosets derived from GP-UPRs synthesised from the aromatic monomer phthalic acid [1] can be reasonably expected to exhibit a high resistance to biodegradation; which was part of the reason for their original choice as a material. An archetypical example of an aliphatic polyester would be poly(glycerol sebacate), which uses the monomers glycerol (triol) and sebacic acid (dicarboxylic acid) to form a branching polyester thermoset on thermal curing via polyesterification [42]. Given their natural biodegradability, in the present work we chose to focus upon aliphatic polyesters.

Another concern associated with GP-UPRs with respect to the UN's 12th SDG is that of a material's provenance. If the underlying chemistry of a polymer is petrochemical in origin, then any carbon liberated from the material at end-of-life (whether by combustion for energy or biodegradation in landfill) will release fossil carbon, contributing via an anthropogenic route towards global climate change. Since GP-UPRs are synthesised from the oil-derived feedstock monomers propylene glycol, maleic acid, and phthalic acid, the resultant thermosets contain endogenic (fossil-derived) carbon [24]. In the present work we aim to avoid these feedstocks and focussed instead upon a bio-based alternative, with inspiration drawn from a chemical system presented in the literature by Alberts, where a combination of glycerol and citric acid were cured to form a bio-based branched aliphatic polyester thermoset [2]. We build upon this research by adopting the chemical system and adapting it to include a third monomer, itaconic acid, which provides a second carboxylic acid with a lower degree of functionality (dicarboxylic acid); this second lower-functionality molecule was expected to encourage the emergence of alternative thermoset networks. Notably, Alberts found that physical properties of the end thermoset poly(glycerol citrate) could be tuned by modifying the monomeric stoichiometries of the prepolymer resin synthesised [2]. Given this, we identified mass loss as a variable of interest during the synthesis of poly(glycerol citrate itaconate) as it is a proxy for polymerised content within a sample, and sought to optimise this by tuning the stoichiometries of the prepolymer resins mixed. Polymerised content could be an indicator of the cross-link density of these samples; a characteristic, which, if maximised, yields improvements in mechanical properties such as ultimate tensile strength, Young's modulus, and glass transition temperature [31].

Poly(glycerol citrate itaconate) is formed from its monomers by polyesterification, a process which is dependent upon variables such as temperature, stoichiometric ratios, and additive presence [2]. Additionally, reaction times for rigid solid thermoset materials are on the order of weeks when uncatalyzed [38] (catalysis commonly involves strong fuming acids which can corrode equipment [35]). Given the high dimensionality and temporally expensive function evaluation exhibited by this family of polyesters, a sample efficient global optimisation technique, namely Bayesian optimisation, was selected to tackle this problem. Alternative global optimisation techniques were considered but ruled out; for example, the hill-climbing class gradient descent algorithm is reliant on derivatives of the objective function which would have been uncertain given the noisy function evaluations associated with physical systems as opposed to simulated ones [36]. The population class evolutionary and particle swarm optimisation algorithms were suitable but avoided in favour of the more sample efficient surrogate class of algorithms. Surrogate class algorithms are variously formulated, but the three major forms are distinguished by the surrogate models they leverage: polynomial response surface model, kriging (also known as gaussian process regression) and radial basis functions [33]. Kriging is preferable to response surface modelling since it makes no assumption of polynomial order, avoiding a priori bias in surrogate models generated. Furthermore, radial basis function-based surrogate models provide no estimates of uncertainty around the fitted function; functions and associated uncertainties derived using kriging provides the means by which Bayesian optimisation effectively balances exploitation and exploration of the parameter space. Hence, for the development of a part-computational and part-laboratory based workflow for the optimisation of mass loss in poly(glycerol citrate itaconate) we decided to use Bayesian optimisation predicated on kriging.

Originally coined in the 1970s by Mockus et al. [28], Bayesian optimisation was later developed further in the 1990s by Jones et al. into a technique called the efficient global optimisation algorithm [21]- this name and technique was thereafter adopted and popularised by the DoE community. More recently, the original name has seen a revival as the technique and several adaptations have been re-discovered and implemented in a range of contemporary communities including robotics [7], bioengineering [3], and machine learning [10]. In materials science, Bayesian optimisation has been used to predict crystalline structures [41], carry out sample-efficient optimisation of density functional theory simulations [4], and multi-objective optimisation of stability and ion-conductivity in materials [22].

This paper begins by detailing the polymeric system considered, experimental means by which samples were manufactured and characterised, as well as the means by which a suitable parameter space was devised. Hereafter, we detail the algorithmic mechanism behind the Bayesian techniques used before further exploring the initial one-shot quasirandom sampling strategy, sequential Bayesian acquisition functions, and one-shot benchmark sampling routines applied during the course of this work. The method concludes with a brief overview of how the physical system was computationally emulated such that Bayesian techniques could be simulated repeatedly to gain statistical insights. The results section is split between elucidation of results associated with the practical optimisation of the physical system, and those which derived from repeated simulation of optimisation routines carried out upon the emulated system. The paper concludes that the Bayesian sequential routines succeeded in improving the mass loss metric from 48% to 72% of the theoretical limit, whilst also statistically proving that the 'probability of improvement' and 'expected improvement' acquisition functions

respectively exhibited a 1.1% and 5.3% probability of discovering values approaching the emulated global optimum.

2 Method

In optimising the properties of a poly(glycerol citrate itaconate) thermoset, interlinking method was developed spanning both practical experimental technique and mathematical framing. Given this method, various computer programs were written spanning the entire experimental process, primary functionality was the facilitation of human-in-the-loop optimisation in a laboratory setting. Programs were written in the Python programming language and developed into modules, whereafter these were united into a single package, "BO4BT", which has been made available within it's own Github repository alongside its dependencies; a .yaml file is also provided which allows an identical Conda environment to be setup, allowing a user to run this code [18].

2.1 Chemical Background

The condensation reaction occurring between carboxyl and hydroxyl groups is an esterification reaction:

$$RCOOH + HOR' \rightarrow RCOOR' + HOH \tag{1}$$

As opposed to the simple esterification shown in Equation 1, the desired polyester thermoset, poly(glycerol citrate itaconate), is the product of polyesterification between multi-functional monomers.

Polyesterification is a form of polycondensation reaction which takes place as follows: an initial ester is formed from two difunctional monomers, where one must at least be a diol and the other a diacid (alternatively, each may possess a carboxyl and a hydroxyl group), producing a molecule of water and an ester with two spare functional groups- these groups can subsequently react with more difunctional monomers causing polymerisation into a linear polyester (See Figure 1). When monomers with a functionality greater than two are introduced into a polyesterification reaction, then side chains can form along the primary backbone of the growing polyester, which causes the development of a branching polymeric structure (See Figure 2). When a side chain reaches another large growing polyester molecule and reacts with it a crosslink is said to have developed, and when widespread crosslinking occurs generating a 3D network of polyester molecules a thermoset polyester has been created. Figure 2 shows how the monomers employed in the synthesis of poly(glycerol citrate itaconate) give rise to this branching thermoset structure.

Importantly, during the polyesterification process water is generated as the main by-product, and hence it is assumed that virtually all mass loss between prepolymer and final cured thermoset would be associated with water evaporation. Given that each molecule of water produced corresponds to an ester bond formed, mass loss can be said to be a good proxy for extent of polymerisation within a sample [38]. Thus, this research identifies mass loss as a metric of interest and sought to maximise this during optimisation of poly(glycerol citrate itaconate) synthesis: increasing the extent of polymerisation and thereby improving the degree of cross-linking and associated strength of these bio-based thermosets.

2.2 Experimental Method

Synthesis of the poly(glycerol citrate itaconate) thermoset was achieved via an uncatalysed polyesterification reaction (See Figure 2). Acid catalysis was avoided to reduce corrosion damage to equipment, this ultimately reduced the rate at which this reaction could be carried out but did not prevent polycondensation [4]. The monomers glycerol (99+%), citric acid (99.6%), and

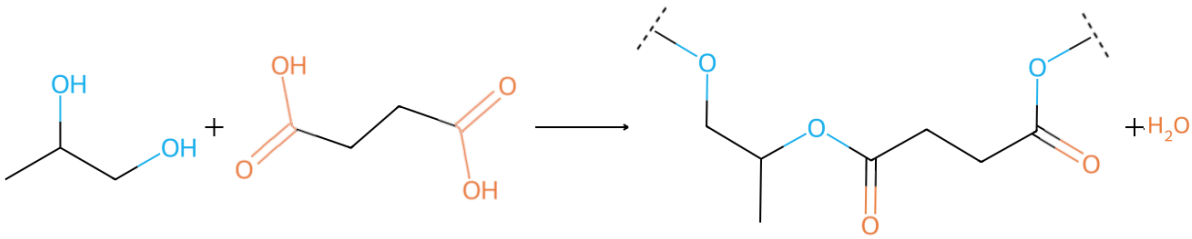

Figure 1: *Archetypical polyesterification between a diol (propylene glycol) and a diacid (succinic acid) forming a linear polyester. Functional groups participating in the reaction are highlighted: hydroxyl groups blue, and carboxyl groups orange. In the products, ester groups are blue and carbonyl groups and water are also orange in accordance with provenance.*

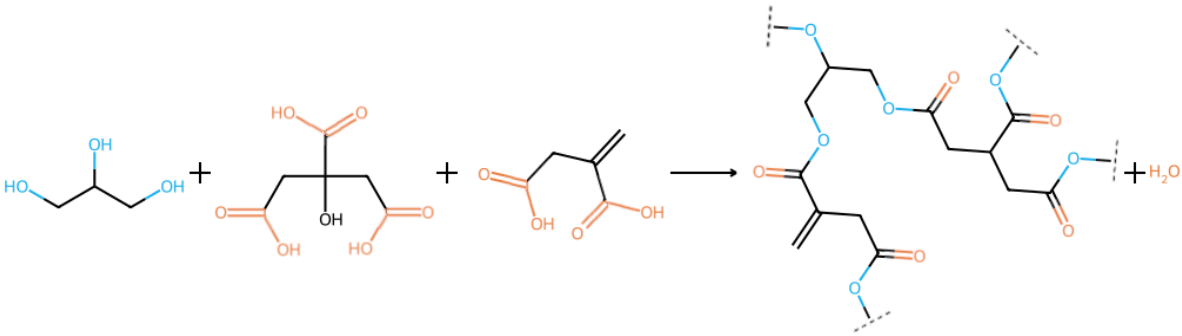

Figure 2: *Polyesterification between glycerol, citric acid, and itaconic acid forming poly(glycerol citrate itaconate). Functional groups participating in the reaction are highlighted: hydroxyl groups blue, and carboxyl groups orange. In the products, ester groups are blue and carbonyl groups and water are also orange in accordance with provenance.*

itaconic acid (99+%) used were supplied by *Fisher Scientific International* and manufactured by *Acros Organics*.

Thermoset synthesis was two-step: firstly, a homogenous prepolymer resin was prepared by stirring solid monomers (acids) together with liquid monomer (glycerol) atop a hotplate and raising the temperature to a maximum of 140 °C until the solids were completely dissolved. Note, 140 °C was adopted to accelerate dissolution without encouraging decarboxylation of the citric acid to itaconic acid [2]. Secondly, prepolymer was cured into the desired thermoset by transferral to a silicone mould before being exposed to elevated temperatures (90 °C) for an extended period (up to two weeks) within a drying oven.

Preliminary experimentation showed that temperatures exceeding 100 °C, caused boiling of byproduct water within samples. Gaseous water from boiling was entrapped within the curing polyester creating bubbles within the final structure, resulting in unpredictable densities of the final material. Given the unpredictable nature of samples generated in this fashion, drying oven temperatures were held below 90 °C to avoid this uncontrollable effect upon the experiments.

As defined by Equation 2, determination of mass loss associated with the synthesis of a thermoset sample (Δm) was obtained by measuring the difference between both the initial mass of prepolymer resin (m_p) and the final mass of cured thermoset (m_t). Measurements were made with a mass balance with a readability of 0.01 g and repeatability of 0.01 g (1σ). Since prepolymer masses weighed 3 g and subsequently recorded mass losses ranged from between 0.3-0.9g, this mass balance was deemed accurate enough for the purposes of this experiment.

$$\Delta m = m_p - m_t \tag{2}$$

Equation 3 defines sample mass loss (Δm) normalised into a percentage format ($\Delta m_{\%}$), this allowed inter-comparison since individual samples had slightly varying initial prepolymer resin masses (m_p).

$$\Delta m_{\%} = \frac{\Delta m \times 100}{m_p} \tag{3}$$

Residence time within the drying oven for a sample was controlled by periodic measurement of sample mass: mass was plotted against time, an exponential decay curve fitted, and the sample removed from the oven once a pre-determined rate of mass loss was reached (this equated to around -0.028% h^{-1} of the initial mass).

2.3 Optimisation Framing

Having selected percentage mass loss within a thermoset sample ($\Delta m_{\%}$) as a suitable target variable for optimisation, predictor variables were defined and constrained creating a suitable parameter space describing a full range of prepolymer monomeric compositions.

The polyesterification of monomers to polymer seen in Figure 2 is presented in Equation 4 as a parameterised stoichiometric equation linking monomeric reagents glycerol (G), itaconic acid (It), and citric acid (Ci) with the product polymer poly(glycerol citrate itaconate) (P(GCiIt)) and byproduct water. Equation 4 introduces two stoichiometric variables representing the moles of glycerol (n_G), citric acid (n_{Ci}) and itaconic acid (n_{It}), and a variable representing the unknown number of moles of water produced (v).

$$\begin{aligned} n_G\ G + n_{Ci}\ Ci + n_{It}\ It \rightarrow \\ 1\ P(GCiIt) + v\ H_2O \end{aligned} \tag{4}$$

The polyesterification reaction can be considered to consist of two possible polymerisation reactions between the triol glycerol and either the triacid citric acid or the diacid itaconic acid. By setting the glycerol stoichiometric variable to unity ($n_G = 1$), Figure 3 shows how the two separate reactions can be considered as if they were stoichiometric ranges, where there exist stoichiometric bounds between which polymerisation is possible.

These stoichiometric bounds are used to directly constrain the stoichiometric variables; Equation 5 describes the lower (n^l_{Ci}) and upper (n^u_{Ci}) bounds placed on the citric acid stoichiometric variable, whilst Equation 6 shows those for lower (n^l_{It}) and upper (n^u_{It}) bounds on proportion of itaconic acid.

$$n^l_{Ci} \leq n_{Ci} \leq n^u_{Ci} = 0.5 \leq n_{Ci} \leq 2.0 \tag{5}$$

$$n^l_{It} \leq n_{It} \leq n^u_{It} = 1.0 \leq n_{It} \leq 2.0 \tag{6}$$

In addition to the basic polymerisable bounds imposed on the two stoichiometric variables, a further constraint must be imposed for a successful optimisation: it must be possible for permutations of prepolymer monomeric mixtures to allow one acid to entirely dominate the other whilst ensuring that a prepolymer will still polymerise- this constraint ensures that any polymerisable stoichiometric mixture of monomers is possible. To achieve this, a three-dimensional bounded domain (a.k.a. parameter space, A) was devised consisting of three fundamental predictor variables: firstly, a citric acid strength predictor variable (s_1) influencing the citric acid stoichiometric variable. Secondly, an itaconic acid strength predictor variable (s_2) influencing the itaconic acid stoichiometric variable. Thirdly, a balancing predictor variable (b_1) capable of trading off the two acid systems, influencing both the citric and itaconic acid stoichiometric variables. $A \subseteq \mathbb{R}^d$ describes the parameter space A as a multi-dimensional real of d dimensions, where $d = 3$ given our three predictor variables. All three predictor variables took constrained ranges of between 0 and 1 ($s_1 = [0, 1], s_2 = [0, 1], b_1 = [0, 1]$), and were linked mathematically to the stoichiometric variables (via equations 7 and 8).

$$n_{Ci} = (n^l_{Ci} + (n^u_{Ci} - n^l_{Ci}) \times s_1) \times b_1 \tag{7}$$

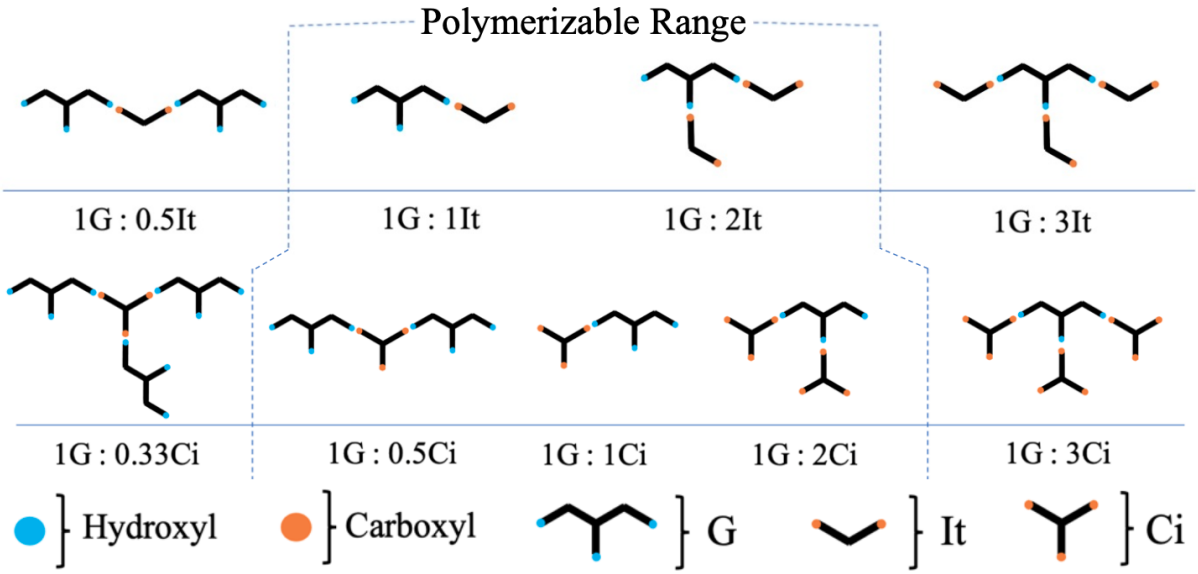

Figure 3: *Simplified chemical representations displaying the polymerisable range of the two different triol-acid combinations encountered during poly(glycerol citrate itaconate) synthesis. In the legend: "G" glycerol, "It" itaconic acid, "Ci" citric acid.*

$$n_{It} = (n^l_{It} + (n^u_{It} - n^l_{It}) \times s_2) \times (1 - b_1) \quad (8)$$

In summary, any random set of predictor variable values drawn from their respective constrained ranges of 0 to 1 (e.g. $s_1 = 0.75, s_2 = 0.25, b_1 = 0.5$) could be used to calculate a unique permutation of stoichiometric variables (e.g. $N_{Ci} = 0.81, N_{It} = 0.63$) which, on mixing of a prepolymer in accordance with Equation 9 and curing, would give rise to a unique polymer and associated mass loss (e.g. $\Delta m_{\%} = 30$).

$$\begin{aligned} n_G\ G + 0 \leq n_{Ci} \leq & n^u_{Ci}\ Ci + \\ 0 \leq n_{It} \leq & n^u_{It}\ It \rightarrow \\ & 1\ P(GCiIt) + v\ H_2O \end{aligned} \quad (9)$$

The sequence of calculations linking abstract predictor variables (s_1, s_2, b_1) to the practical stoichiometric variables (n_{Ci}, n_{It}) and onwards to the target variable of mass loss $(\Delta m_{\%})$, allows the expression of a black-box mapping between predictors and targets. This black-box mapping is described by $f : A \rightarrow \mathbb{R}$, where an unknown function (f) maps any point in our parameter space (A) to a single real number value $(\mathbb{R})$. Equation 10 more explicitly expresses the black-box function in terms of the our target variable and predictor variables at a given sample number (i). Note that sample number is an integer counter used to track the number of function evaluations carried out during a sequential optimisation.

$$\Delta m_{\%} = f(\mathbf{x}) = f([s^i_1, s^i_2, b^i_1]) \quad (10)$$

Our optimisation problem is thus expressed in Equation 11 as the maximisation of the black-box function, where the global optimum set of predictor variables $(\mathbf{x}^*)$ is sought.

$$\mathbf{x}^* = \arg\max_{\mathbf{x} \in A} f(\mathbf{x}) \quad (11)$$

2.4 Optimisation Algorithm

The sequential optimisation algorithm employed during this work is summarised by the flow diagram in Figure 4. Succesful conclusion of the optimisation is controlled by an if-else clause comparing an initially specified permissable number of function evaluations (a.k.a. sample limit, p_{tot}) with the current sample number (i): when $i < p_{tot}$ the algorithm continues. A second if-else statement controls whether a dataset exists from which a Gaussian process prior might be built for Bayesian sampling. When available, the algorithm uses the Gaussian process to draw a pre-defined number of samples (p_{iter})

from the parameter space for evaluation. If the prior-forming dataset is lacking, an initial quasirandom (Sobol) sampling of the parameter space with a pre-defined number of deterministic samples (p_{init}) is initiated. Samples drawn from the parameter space are processed into stoichiometric variables (n_{Ci}, n_{It}), with which an experimenter can manufacture sample(s) before evaluating the target variables $\Delta m_{\%}$ value(s) and returning these to the algorithm.

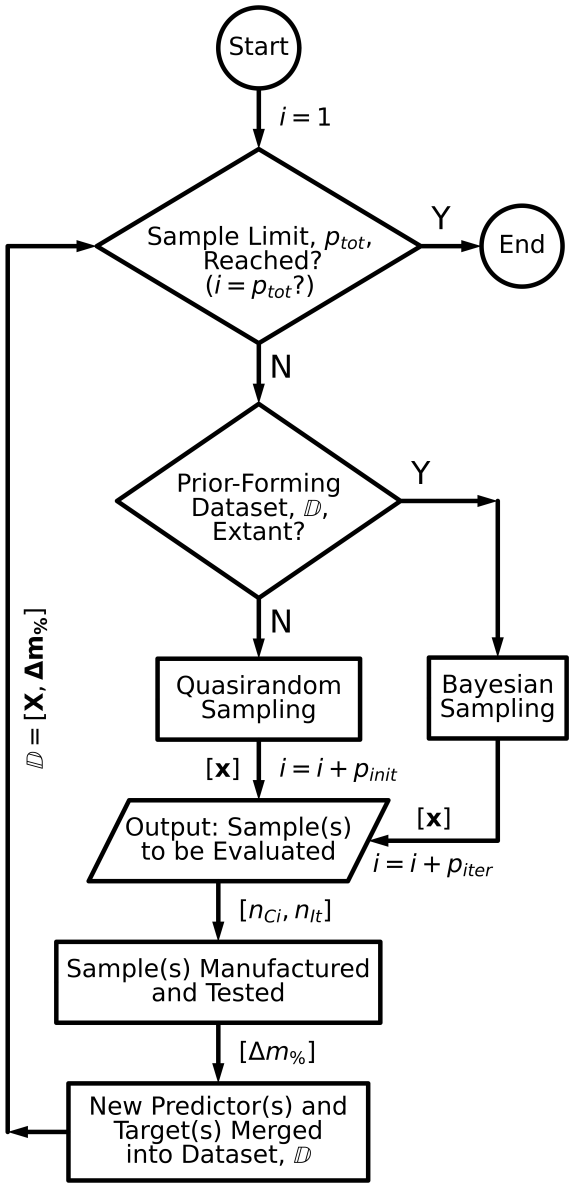

Figure 4: *Flow diagram depicting the sequential optimisation algorithm developed.*

The algorithm seen in Figure 4 was implemented in the Python programming language, with multiple modules written and united into a single package with the acronym "BO4BT" (version 1.0.0)- "Bayesian optimisation for biodegradable thermosets" [18].

On calculation of stoichiometric variables during an iteration, the algorithm merges all new sets of predictors variable values into a growing predictor-matrix ($\mathbf{X}$), shown in Equation 12.

$$\begin{aligned}\mathbf{X} =& [\mathbf{x}^{i=1}, \mathbf{x}^{i=2}, ...] \\ =& [[s_1^{i=1}, s_2^{i=1}, b_1^{i=1}], [s_1^{i=2}, ...], ...]\end{aligned} \tag{12}$$

On conclusion of an iteration, the algorithm merges all data associated with samples evaluated into an optimisation-governing dataset. Equation 13 presents this dataset ($\mathbb{D}$) as a matrix which consists of the predictor-matrix ($\mathbf{X}$) and an array of target

variable values ($\mathbf{\Delta m}_{\%}$).

$$\begin{aligned}\mathbb{D} =&[\mathbf{X}, f(\mathbf{X})]\\ =&[\mathbf{X}, \mathbf{\Delta m}_{\%}]\\ =&[[\mathbf{x}^{i=1}, \mathbf{x}^{i=2}, ...], [\Delta m_{\%}^{i=1}, ...]]\end{aligned} \tag{13}$$

2.5 Sampling Procedures

During a sequential optimisation two forms of sampling were employed: firstly, a deterministic space-filling sampling of a quasirandom nature was used to representatively sample the parameter space when initialising the optimisation. Subequently, a Bayesian sampling technique was used to sample-efficiently guide the optimisation's search for the global optimimum x^*.

2.5.1 Initial Quasirandom Sampling

Optimisation routines were initiated by representatively sampling across the parameter space A; achieved using the quasirandom Sobol sampling algorithm to draw a preset number (p_{init}) of samples from across A. Sobol sampling was employed as it provides the same or better space-filling properties across $\mathbb{R}^{1\leq d\leq 40}$ domains as compared with pseudorandom or grid techniques [17]. On evaluation of the initial representative set of samples ($[\mathbf{x}^{i=1}, ..., \mathbf{x}^{i=p_{init}}]$) for the target variable values ($[\Delta m_{\%}^{i=1}, ..., \Delta m_{\%}^{i=p_{init}}]$), an initial prior-forming dataset ($\mathbb{D}^{i=p_{init}}$) is produced.

In practice, only one quasirandomly sampled initial prior-forming dataset ($\mathbb{D}^{i=8}$) was generated and used (visualised in Figure 5); this allowed a fair inter-comparison of the two Bayesian optimisation routines attempted.

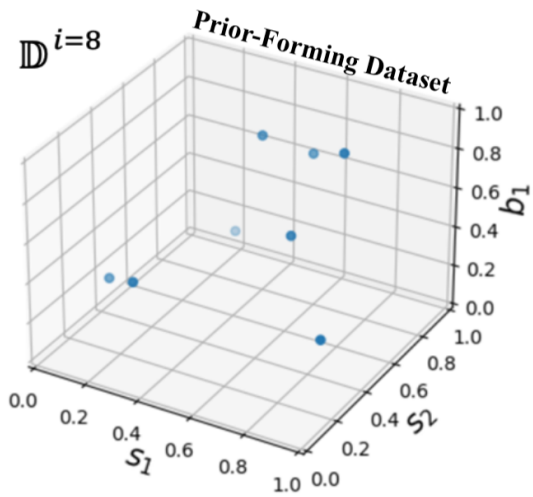

Figure 5: *This 3D graphic depicts the initial 8 samples quasirandomly drawn from across the parameter space, A, comprised of s_1, s_2, and b_1 (as in equations 7 and 8), which provide the prior-initialising dataset for the Bayesion routines. Coloration of points denote distance from viewer with darker points appearing further away.*

2.5.2 Sequential Bayesian Sampling

Sequential sampling used in this research abided by Bayes theorem (see Equation 14), where a posterior probability $P(H|E)$ (updated probability of hypothesis H given new evidence E) is equal to the prior probability (original probability of hypothesis) multiplied by the likelihood $P(E|H)$ (probability of evidence given hypothesis) divided by the marginal probability $P(E)$ (probability of evidence). As new evidence becomes available, the posterior probability is recycled into the equation as a prior alongside the new evidence and updated to a new posterior [14].

$$\overset{\text{posterior}}{P(H|E)} = \frac{\overset{\text{prior}}{P(H)} \times \overset{\text{likelihood}}{P(E|H)}}{\underset{\text{marginal}}{P(E)}} \tag{14}$$

Bayesian optimisation uses a Bayesian statistical model called a Gaussian process regression model (a form of Bayesian posterior distribution) derived from Gaussian process regression of a dataset $\mathbb{D}$ [32]. Equation 15 shows how a gaussian process, $\mathcal{GP}$,

is a distribution over functions which is completely described by both a mean function, m, and a covariance function, K.

$$f(\mathbf{x}) \sim \mathcal{GP}(m(\mathbf{x}), K(\mathbf{x}, \mathbf{x}')) \tag{15}$$

More intuitively, Equation 16 describes the gaussian process as a surrogate function $\hat{f}(\cdot)$ describing the link between a candidate array of predictor values $\mathbf{x}$ and a tuple containing the mean and the variance of a normal distribution over values of f at x.

$$\hat{f}(\cdot) : \mathbf{x} \mapsto [m(\mathbf{x}), K(\mathbf{x}, \mathbf{x}')] \tag{16}$$

As an infinite-dimensional generalisation of multivariate normal distributions the gaussian process inherits the properties of a conjugate prior [13], allowing the initial posterior (e.g. $\mathcal{GP}^{i=8}$) to be recycled as a prior alongside new data (e.g. $\mathbb{D} + [\mathbf{x}^{i=9}, \Delta m^{i=9}]$) to generate an updated posterior (e.g. $\mathcal{GP}^{i=9}$) in accordance with Bayes theorem (see 14). The surrogate model therewith evolves during the course of sequential optimisation as the number of iterations, i, increases: corresponding to the updating of beliefs when considering new evidence which hallmarks all Bayesian techniques.

During fitting of the Gaussian process a covariance function is used to encode our beliefs regarding the families of functions likely to fit the data observed; this work made use of the Matérn family. When using the Matérn covariance function a hyperparameter controlling smoothness of functions drawn from the Gaussian process was set ($\nu = 2.5$)- this encouraged the generation of relatively smooth functions, which was deemed acceptable given that preliminary sampling of the parameter space indicated a similar topography.

Subsequent of posterior genesis, a simple one-step lookahead sampling policy was used to select a suitable next point $\mathbf{x}$ within the parameter space A to sample. This research focussed upon policies that made use of the 'simple reward' utility function (see Equation 17) which takes the current dataset $\mathbb{D}$, and from the set of sample coordinates evaluated so far X returns the predicted mean value $\Delta m_{\%}$ of the point $\mathbf{x}$ exhibiting the highest predicted mean ($max\ m(X)$) according to the current posterior $\mathcal{GP}$ [16].

$$u(\mathbb{D}) = max\ m(X) \tag{17}$$

In adopting this utility function, two different sampling policies were developed and investigated: the first was predicated on the 'probability of improvement' acquisition function (see Equation 18) where any given putative $\mathbf{x}$ is scored by its probability that it will produce an updated dataset $\mathbb{D}'$ that will generate a higher evaluation from the utility function than an absolute threshold τ. Probability of a sample exceeding the current best is deduced using the current posterior $\mathcal{GP}$'s mean ($m(\mathbf{x})$)and standard deviation ($\sqrt{K(\mathbf{x}, \mathbf{x})}$) measured at that sample position. Note, this work made the absolute threshold equal to an evaluation of the utility function using the current dataset ($\tau = u(\mathbb{D})$), eschewing the added complication of deciding upon a marginal threshold parameter ε which could have been used ($\tau = u(\mathbb{D}) + \varepsilon$) to offset any overly exploitative behaviour induced by rewarding any improvement no matter the scale.

$$\begin{aligned} \alpha_{PI}(\mathbf{x}) =& Pr(u(\mathbb{D}') > \tau) \\ =& \int_{\tau}^{\infty} \mathcal{N}(m(\mathbf{x}), \sqrt{K(\mathbf{x}, \mathbf{x})}) dm(\mathbf{x}) \\ =& 1 - \Phi((\tau - m(\mathbf{x}))/\sqrt{K(\mathbf{x}, \mathbf{x})}) \end{aligned} \tag{18}$$

The second policy made use of the 'expected improvement' acquisition function (see Equation 19), which incorporates an expectation of improvement value ($u(\mathbb{D}') - u(\mathbb{D})$); this counteracts overtly exploitative behaviour by rewarding both the scale and probability of an improvement above the threshold τ [27].

$$\begin{aligned} &\alpha_{EI}(\mathbf{x}) = \int_{\tau}^{\infty} c\, \mathcal{N}(m(\mathbf{x}), \sqrt{K(\mathbf{x}, \mathbf{x})}) dm(x) \\ &\text{where } c = u(\mathbb{D}') - u(\mathbb{D}) \end{aligned} \tag{19}$$

Equation 20 shows how the next point to be sampled sampled $\mathbf{x}^+$ is found by trivially optimising the cheap acquisition function (many putative samples $\mathbf{x}$ may be evaluated) to find the global optimum within the bounds of the parameter space A [27].

$$\mathbf{x}^+ = \arg\max_{\mathbf{x}\in A} \alpha(\mathbf{x}) \tag{20}$$

Gaussian processes and acquisition functions were produced using two open-source packages: scikit-optimise ("skopt", v0.9.0) [23] and Ax ("ax-platform", v0.4.3) [15].

2.6 Benchmarks

Sample efficiency of the Bayesian sequential optimisation techniques was proven by employing three simple benchmarks. These were derived using one-shot deterministic space-filling sampling techniques including pseudorandom, quasirandom (Sobol), and grid to disperse the maximum allowable number of samples ($p_{tot} = 27$) across the parameter space A (See Figure 6). Subsequently, for each of the techniques deployed, all samples $\mathbf{X}$ were evaluated $f(\mathbf{X})$ to retrieve associated target variable values $\mathbf{\Delta m}_{\%}$ generating a complete dataset $\mathbb{D}$. The highest target variable value was identified within each technique's dataset ($\Delta m^{\circ}_{\%}$) and equated to the benchmark value (See Table 1).

Benchmark Technique	Max (%)	Mean (%)	Standard Dev. (%)
Grid	16.24	11.93	1.66
Quasirandom	14.55	12.34	1.43
Pseudorandom	14.01	12.54	0.88

Table 1: ***BenchmarkBestValues*** *Statistical analysis of results for 27 samples derived via one-shot deterministic sampling techniques.*

2.7 Emulation of Physical System

An emulation of the physical system investigated facilitated statistical analyses of the various sampling techniques' efficacies in discovery of the experimental global optimum. This emulation was developed by firstly concatenating the final datasets $\mathbb{D}$ generated from the three benchmark and two sequential techniques deployed- this master dataset which represented all the knowledge of the objective function's topography within the parameter space investigated. Subsequently, a Gaussian process $\mathcal{GP}$ was trained on this master dataset using the parameterisations described earlier- the resultant mean function $m(\mathbf{x})$ then being used as a cheap surrogate function of the latent objective function (i.e. the emulator). Note, that some data obtained outside the sample limit was included in this master dataset- specifically, the final samples generated by the Bayesian sequential technique using the 'probability of improvement' acquisition function; these were remarkably high, and represented the experimental global optimum discovered during the course of this work. Care was taken to ensure the trained $\mathcal{GP}$ faithfully replicated this region of optimality within the parameter space as well as the general trends of the objective function.

3 Results

This work produced two sets of results: firstly, those derived from optimisation conducted via sampling directly from the latent objective function described by the physical system. Secondly, those stemming from the optimisation of a surrogate function trained to emulate the latent objective function and the associated physical system. The latter set of results were generated to provide statistical support for the former.

3.1 Physical System Optimisation

Two Bayesian sequential optimisation experiments were run, each with the same sample limit ($p_{tot} = 27$), number of samples drawn

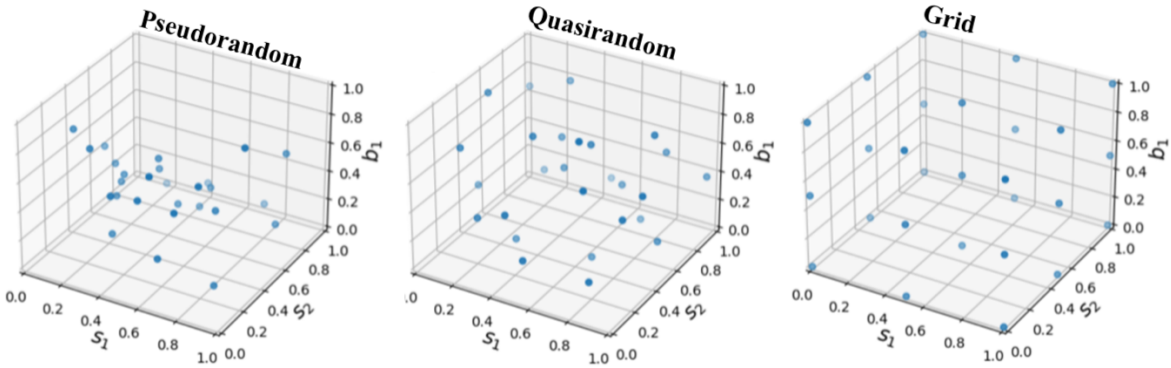

Figure 6: *Deterministic space-filling one-shot sampling techniques. Coloration of points denote distance from viewer with darker points appearing further away.*

per Bayesian iteration ($p_{iter} = 3$), and prior-forming dataset of 8 quasirandomly sampled targets (See Figure 5). Differing between these two methods was the acquisition function employed: the first using the 'probability of improvement' whilst the second used the 'expected improvement' function. Figure 7 shows the optimisation progression of these two sequential techniques when compared with the one-shot deterministic methods; mass losses plotted on the y-axis are the best values uncovered given the samples evaluated so far, $\Delta m^{\circ}_{\%}$. A statistical overview is presented in Table 2 which concerns the 8 quasirandom and 16 sequential samples for each of the two Bayesian techniques. Note, that given the sequential samples available ($p_{tot} - p_{init} = 19$) and the samples per iteration ($p_{iter} = 3$), an extra iteration of sequential sampling was run, which breached the sample limit- these extra samples are included in Figure 7, but are not considered in the preparation of Table 2.

By comparing Tables 2 and 1 it is apparent that both sequential routines outperformed the one-shot techniques in terms of best sample discovered. Figure 7 shows that the sequential Bayesian technique employing the 'expected improvement' acquisition function discovered a point exceeding the strongest one-shot technique in almost half the allowable samples (63% p_{tot}). In contrast, the 'probability of improvement' acquisition function only managed to discover a point surpassing the grid sampling benchmark at the sample limit itself (100% p_{tot})- this hinted at the 'probability of improvement' acquisition function exhibiting more exploratory behaviour than the 'expected improvement' acquisition function. Further evidence of the 'probability of improvement' acquisition functions exploratory bias is seen in Table 2, where the samples generated by this sequential routine obtain a markedly lower mean than that of those drawn by the 'expected improvement' acquisition function; anecdotally, on reviewing the placement of samples in 3D space it was noted that these were generally placed in exploratory locations within the parameter space. Interestingly, the evidence collated above allayed fears that the 'probability of improvement' acquisition function would exhibit overtly exploitative behaviour due to the lack of a marginal threshold parameter ε being manually tuned.

Acquisition Function	Max (%)	Mean (%)	Standard Dev. (%)
PI	16.67	12.50	1.63
EI	17.99	13.67	1.91

Table 2: ***BayesianSequentialsBestValues*** *Statistical analysis of results for samples derived via sequential Bayesian sampling techniques. EI denotes 'expected improvement', whilst PI intends 'probability of improvement' acquisition function.*

Notably, the Bayesian sequential technique employing the 'probability of improvement' acquisition function discovered samples attaining the highest mass losses of

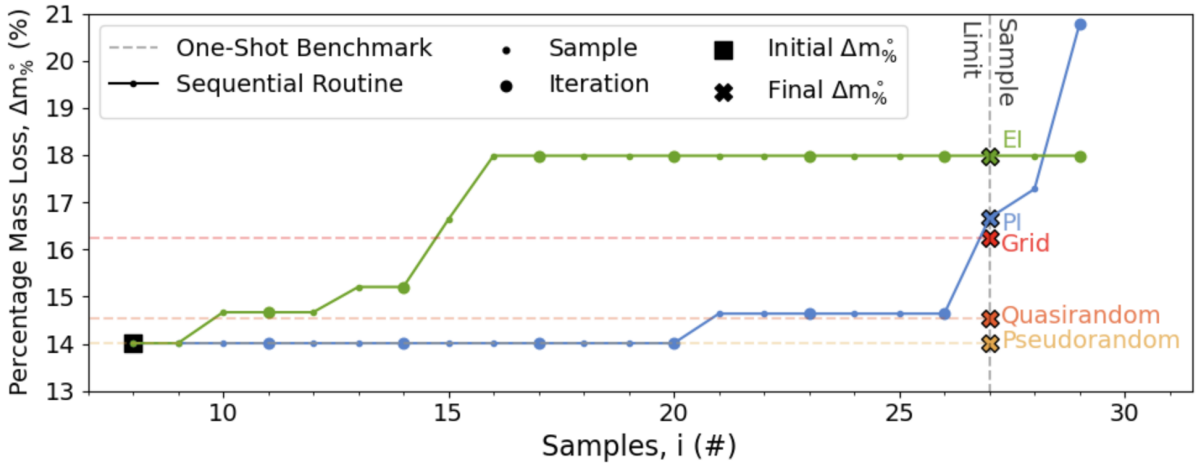

Figure 7: *A plot describing the optimisation progression for the two Bayesian sequential techniques deployed. EI denotes 'expected improvement', whilst PI intends 'probability of improvement' acquisition function.*

any sample characterised during this work ($\Delta m^{\circ}_{\%} = 20.77\%$), albeit outside the sample limit (See Figure 7).

3.2 Emulated System Optimisation

Given the temporal expense of sampling the objective function, repetition of sequential sampling techniques was infeasible. This precluded statistically proving the utility of Bayesian optimisation in this field within reasonable timeframes. By developing a cheap surrogate function capable of emulating the physical system, a much larger number of function evaluations could be carried out computationally, facilitating statistical analysis of sampling methods' efficacy. By despatching 1,000 simulations of each of the two random benchmark one-shot techniques and the two Bayesian sequential techniques, and returning the best target variable values $\Delta m^{\circ}_{\%}$ discovered by each of these, basic statistical conclusions could be drawn. Note that each simulation of the sequential techniques was initiated with a separate quasirandomly drawn prior of 8 samples and that the grid sampler was precluded since its outcomes were not affected by chance.

Figure 8 presents a swarmplot showing dispersion of outcomes for the 4,000 simulations across the four sampling techniques studied. On comparison of the mean $\Delta m^{\circ}_{\%}$ values discovered by the various sampling techniques, sequential Bayesian techniques were shown to generally perform better (PI=14.6%, EI=15.4% for $\Delta m^{\circ}_{\%}$) than one-shot techniques (PR=14.0%, QR=14.1% for $\Delta m^{\circ}_{\%}$). The Bayesian sequential technique adopting the 'expected improvement' acquisition function in its sampling policy not only outperformed the 'probability of improvement' technique in terms of mean value attained, but also in number of simulations obtaining $\Delta m^{\circ}_{\%}$ values which resided within the small region of global optimality. This region was characterised as containing function evaluations exceeding 18.5%, and this simple threshold was therewith used to calculate the number of simulations attaining such globally optimum values ($\Delta m^{\circ}_{\%} > 18.5$) for each of the sampling techniques (PI=11, EI=53 simulations where $\Delta m^{\circ}_{\%} > 18.5$). Notably, neither of the one-shot techniques was able to uncover the globally optimum region (PR=0, QR=0 simulations where $\Delta m^{\circ}_{\%} > 18.5$). Given the above, and a (particularly) tentative assumption that the final emulation was an accurate description of the underlying objective function, we could venture that the probability of having discovered the experimental global optimum using the 'expected improvement' and 'probability of improvement' techniques was 5.3%

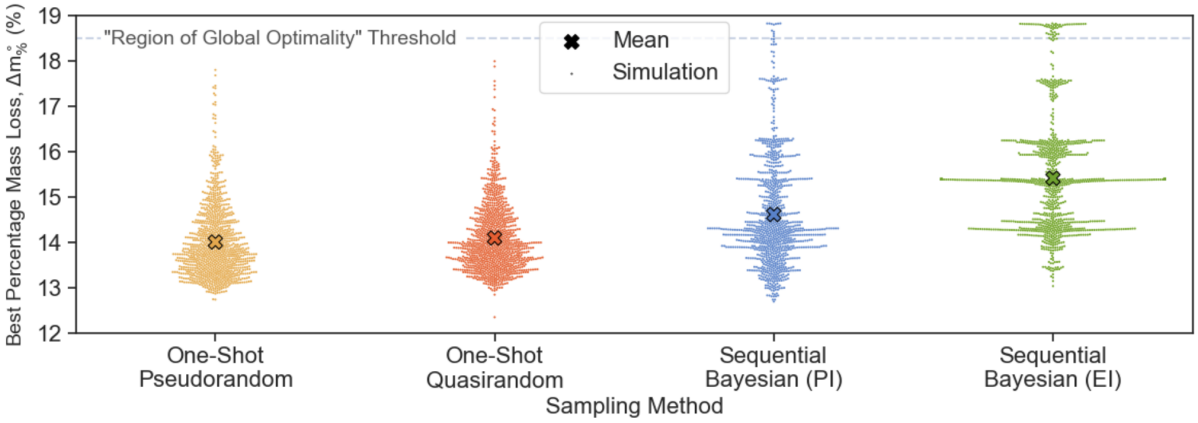

Figure 8: *Swarm plot showing the best function evaluations derived from each of 1,000 simulations run for each sampling technique predicated on initial random sampling.*

and 1.1% respectively. This would make the practical reality of having discovered these points remote but most definitely not impossible; and, importantly, the probabilities are a marked improvement above the 0.0% exhibited by the one-shot techniques.

4 Conclusion

Having engaged in the practical optimisation of the monomeric composition of a simple bio-based branching thermoset polyester (poly (glycerol citrate itaconate)) with respect to an objective of interest (percentage mass loss via curing) we have discovered an experimental global optimum of $\Delta m^{\circ}_{\%} = 20.77\%$ at the sample coordinates ($s_1 = 0.337$,$s_2 = 0.648$,$b_1 = 0.773$) corresponding to a stoichiometric balance of 1 glycerol to 0.77 citric acid to 0.37 itaconic acid; this being discovered by a sequential Bayesian technique whose sampling policy adopted the well known 'probability of improvement' acquisition function. Furthermore, this sequential Bayesian technique and a second (which employed the alternative 'expected improvement' acquisition function) displayed better sample-efficiency than simple one-shot techniques in the field.

Utility of these Bayesian techniques in sample-efficient global optimisation was proven through 1,000 simulations of their behaviour when sampling a surrogate function emulating the objective function; the 'expected improvement' and 'probability of improvement' sequential techniques were found to exhibit a 5.3% and 1.1% probability of uncovering the experimental region of global optimality, whilst both the pseudorandom and quasirandom one-shot techniques entirely failed to sample these regions.

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